

Name: Adalynn Cadence Everly

ID: 1001852753

Class: CSE-4321-001

HW 8:

CSE 4321, HW 8 Topic: Regression Testing Note that for the following problems, you must show intermediate steps in order to get full credits.

1. (20 points) Suppose program P has been executed against a test suite T consisting of six tests, t1, t2, t3, t4, t5 and t6. A total of six entities are covered by the tests as shown in the following table: 0 (or 1) in a column indicates that the corresponding entity is not covered (or covered). The entities could be basic blocks in the program, functions, def-uses, or any other testable element of interest. Follow procedure CMIMX to find the minimal cover set for the six entities.

	1 2 3 4 5 6	

t1	0 1 0 1 0 1	
t2	1 0 0 0 1 0	
t3	1 1 0 1 0 0	
t4	1 0 1 0 0 0	
t5	0 1 0 1 1 0	
t6	1 0 0 0 0 0	

HW 8

① Let's track uncovered entities $\{1, 2, 3, 4, 5, 6\}$
Test T3 covers $\{1, 2, 4\}$ → 3 entities.

Select T3

Covered so far: $\{1, 2, 4\}$

Remaining: $\{3, 5, 6\}$

Step 2: pick test that covers most of what's still covered.

T1 $\rightarrow \{2, 4, 6\}$ → Only E6 is new → 1 entity

T2 $\rightarrow \{1, 5\}$ → E5 = new

~~T3 $\rightarrow \{1, 2, 4\}$ → E1, E2, E4 = new~~ → 3 entities

T4 $\rightarrow \{1, 3\}$ → E3 = new

T5 $\rightarrow \{2, 4, 5\}$ → E5 = new

T6 $\rightarrow \{1\}$ → all already covered → 0

Tie: T1 (gets E6), T2 (gets E5), T4 (gets E3)

Pick T1 arbitrarily tie breaker

Covered so far: $\{1, 2, 4, 6\}$

Remaining: $\{3, 5\}$

Step 3: again, pick test that adds the most new coverage

Now remaining: $\{3, 5\}$

T2 $\rightarrow$ gets E5

T4 $\rightarrow$ gets E3

T5 $\rightarrow$ gets E3

T6 $\rightarrow$ no hit

Pick T4 (gets E3)

Covered so far: $\{1, 2, 3, 4, 6\}$

Remaining: $\{5\}$

Step 4: pick test that covers E5

T2 $\rightarrow$ gets E5

T5 $\rightarrow$ gets E5

Pick T2 arbitrarily

Final covered set: $\{1, 2, 3, 4, 5, 6\}$

min covered set: $\{T3, T1, T4, T2\}$

2. (20 points) Suppose that there is an application P consisting of 8 methods, $m_1 \dots m_8$. Also suppose that there is a regression-test set $T = \{t_1, t_2, t_3, t_4, t_5\}$. The methods covered by each test in T are listed in the following table. Follow procedure PrTest to obtain a prioritized list of tests based on residual coverage.

Test (t)	Methods covered (cov(t))	cov(t)

t1	m_1, m_3, m_5, m_6, m_8	5
t2	m_1, m_7, m_8	3
t3	m_1, m_2, m_3, m_5	4
t4	m_1, m_2, m_3, m_4	4
t5	m_1, m_5, m_8	3

Initial Coverage:

universe of methods: $\{m_1, m_2, m_3, m_4, m_5, m_6, m_7, m_8\}$
 initial residual = all 8 methods
 we pick the test that covers the most new methods at each step.

Step 1:

Check how many unique methods each test covers.

T1: $\{m_1, m_3, m_5, m_6, m_8\} \rightarrow 5$

T2: $\{m_2, m_3, m_5, m_6, m_8\} \rightarrow 5$

T3: $\{m_2, m_7, m_8\} \rightarrow 3$

T4: $\{m_1, m_2, m_3, m_5\} \rightarrow 4$

T5: $\{m_1, m_5, m_8\} \rightarrow 3$

Pick t1 (covers 5 new methods) $\rightarrow$ select t1

res: $\{m_2, m_4, m_7\}$

Step 2: recalc residual coverage

T2: $\{m_7\} \rightarrow 1$

T3: $\{m_2\} \rightarrow 1$

T4: $\{m_2, m_4\} \rightarrow 2$

T5: $\{m_3\} \rightarrow 1$

Pick t4

res: $\{m_7\}$

Step 3: only test covering m_7 is t2 $\rightarrow$ Pick it

fin prioritized test list

$\{t1, t4, t2, t3, t5\}$

(NOTE: t3 & t5 are now redundant but added @ the end to complete list.)